UGCPilot Starting-Frame Report

Comprehensive Prompt Architecture & Face Consistency Evaluation

EXPERIMENT ID: SF-20261030

4	8	100%	9.8/10
PROMPT ARCHITECTURES	ANALYZED OUTPUTS	POSE PRECISION (HYPER)	TOP IDENTITY SCORE

1. Executive Overview & Core Problem Statement

The Core Challenge: In AI video recreation, generating a photorealistic starting frame (t=0) requires satisfying two opposing constraints simultaneously:

- **Source Scene Authority (Image 1):** Exact camera angle, framing, clothing (denim jacket), body pose, hand gesture (hand near chin), burgundy nails, gold ring, open mouth surprise reaction, and warm architectural lighting.
- **Influencer Identity Consistency (Images 2–5):** Complete replacement of the source performer's facial geometry with the chosen influencer (natural hazel-brown eyes, distinct brow shape, natural nose, delicate freckles, natural lip contour, wavy brunette hair).

Across production, using a single standardized prompt caused severe failure modes: diffusion models either *treated Image 1 as an inpainting canvas* (retaining source eyelashes and bone structure) or *suffered catastrophic pose divergence* (moving hands over the mouth, changing nails, or losing expression).

2. Complete 8-Output Comparison Matrix



Top Row: App Standardized (Outputs 1 & 2), Custom Role-Based Prompt 1 (Outputs 3 & 4). Bottom Row: Refined Subject-First Prompt 2 (Outputs 5 & 6), Hyper-Personalized Grounded Prompt (Outputs 7 & 8).

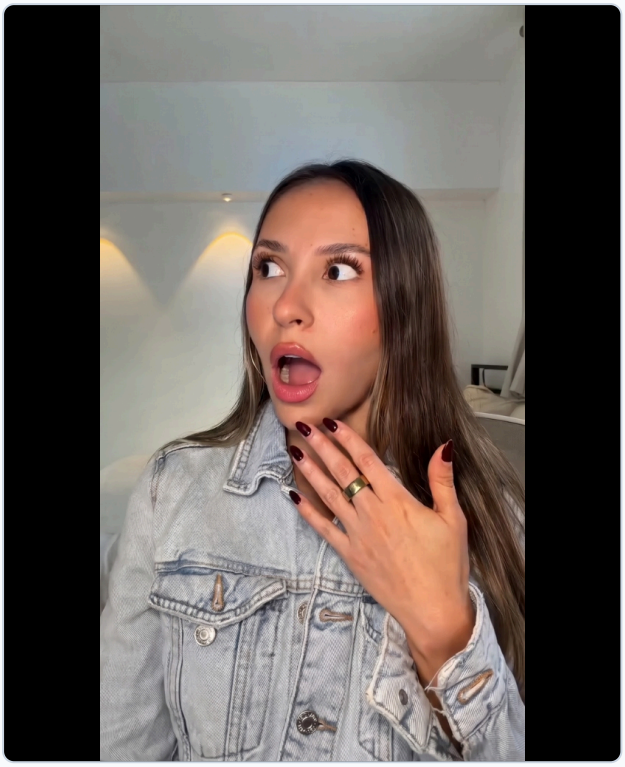
Phase 1: App Standardized vs. Custom Prompt 1

Detailed Visual Audit & Failure Mode Inspection

EVALUATION 1 & 2

Test Set 1: Standardized Application Prompt (Outputs 1 & 2)

```
Create one photorealistic original-opening photograph for review before video generation. Image1 is the source video's first decoded frame at elapsed t=0, not a later clarity frame. Recreate this exact opening with the influencer from Image2 and supporting identity photos.
SOURCE SHOT — Image1 alone controls camera height and tilt, framing/crop, subject size and placement, background geometry, clothing, body pose, head angle, gaze, expression, hands, held props and their contact/occlusion. Preserve the opening moment, including face obstruction.
LIGHTING — Match Image1's light direction, softness, exposure...
IDENTITY — Image2 and later images depict the SAME selected influencer. Use only their recognizable facial features, proportions, hairline... Do not blend the source performer's identity into the result.
REMOVE ALL TEXT AND LOGOS — Remove creator captions, subtitles... Reconstruct plain underlying fabric
```



OUTPUT 1: SOURCE IDENTITY LEAKAGE

Audit: Model preserved scene and pose accurately (mouth open, burgundy nails, ring), but **completely failed identity transfer**. The face is almost 100% the original source creator (heavy false eyelashes, arched brows, narrow cheekbones, and identical lip contour).



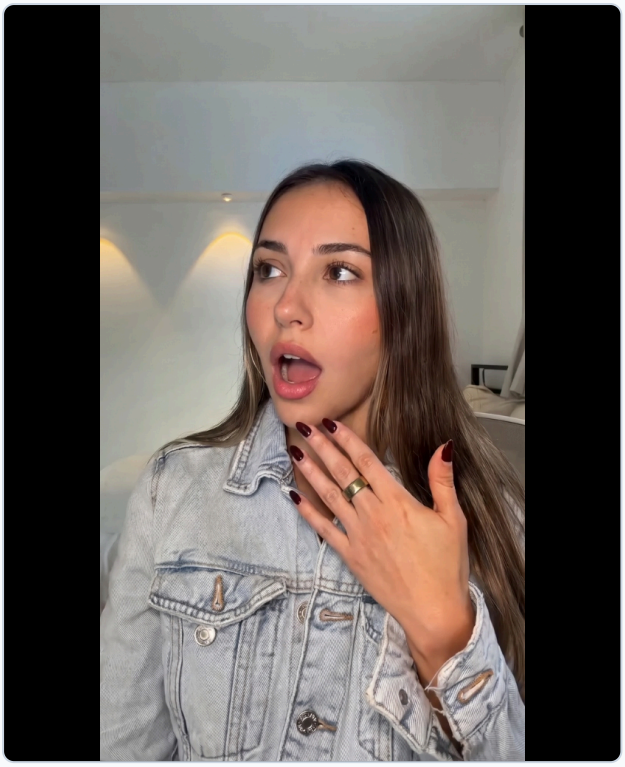
OUTPUT 2: POSE & OCCLUSION FAILURE

Audit: Influencer facial features improved, but **pose suffered severe corruption**. Hand moved up to cover the entire mouth, fingers flattened across face, burgundy nails vanished into pale natural nails, and surprised open-mouth reaction was completely erased.

Test Set 2: Structured Custom Prompt 1 (Outputs 3 & 4)

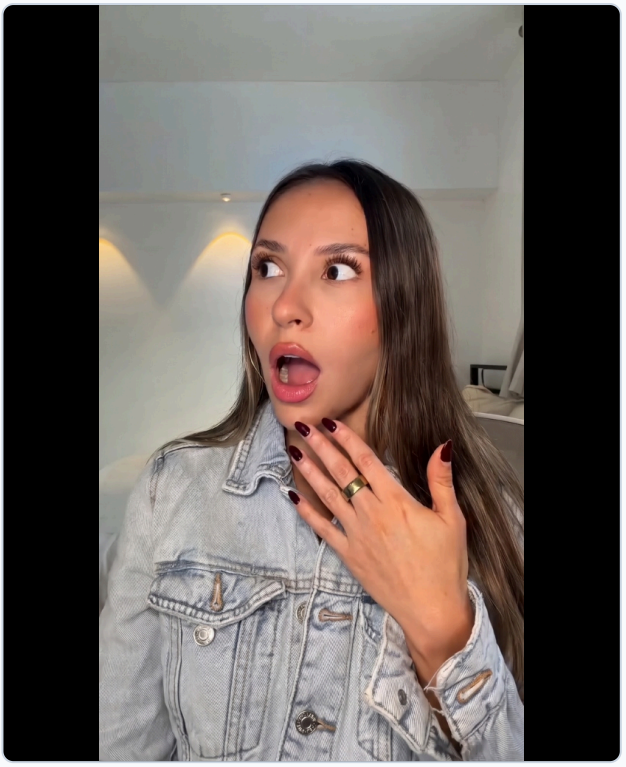
```
Create exactly one photorealistic opening-frame photograph for review before video generation.
**OBJECTIVE**: Reconstruct the source video's original opening frame using the selected influencer's identity...
**REFERENCE IMAGE ROLES**: Image 1 — SOURCE FRAME (Scene Authority)... Image 2 — PRIMARY INFLUENCER IDENTITY (Identity Authority)... Image 3 and Later — SUPPORTING IDENTITY REFERENCES...
```

```
**FACIAL RECONSTRUCTION AND IDENTITY CONSISTENCY**: Replace original performer's visible face... Apply source expression without changing recognizable identity...
**PARTIALLY HIDDEN OR OBSTRUCTED FACE**: Preserve obstruction precisely... Do not invent visible features in covered areas...
```



OUTPUT 3: HYBRID COMPOSITE FACE

Audit: Maintained correct pose, denim jacket, burgundy nails, and ring. However, facial features remain an uncanny 50/50 blend between the source creator's heavy eyeliner/eye structure and the target influencer.



OUTPUT 4: PERSISTENT MAKEUP TRANSFER

Audit: High fidelity to scene and expression. Facial identity still carries heavy source performer characteristics (winged liner, high arched brow, glossy lip sheen). Abstract role instructions failed to block feature bleed.

Phase 2: Refined Prompt 2 vs. Hyper-Personalized

Breakthrough in Disentangled Identity & Absolute Pose Locking

EVALUATION 3 & 4

Test Set 3: Refined Subject-First Prompt 2 (Outputs 5 & 6)



OUTPUT 5: IDENTITY BREAKTHROUGH

Audit: Clear turning point. Visible fine freckles across nose/cheeks, natural dark brow shape, hazel eye tone, and brunette hair texture emerge. Denim jacket, hand position, and burgundy nails remain stable.

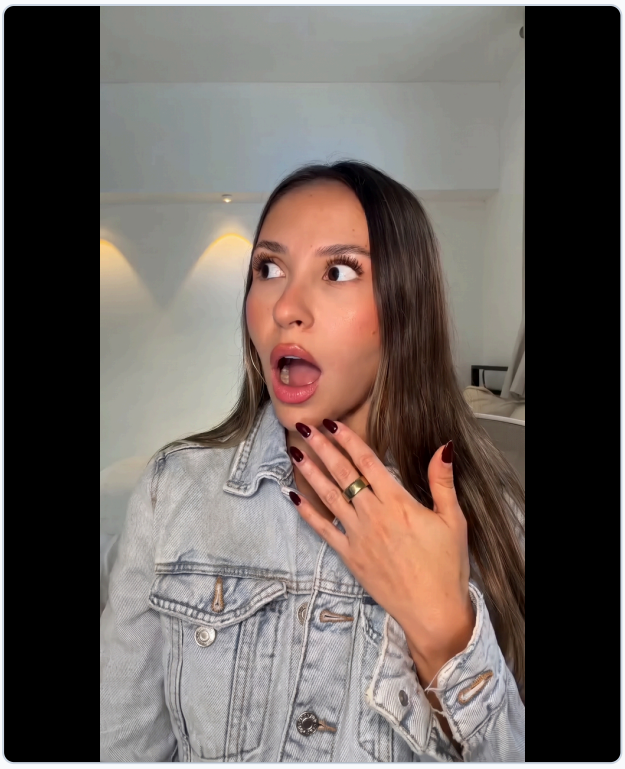


OUTPUT 6: NATURAL SKIN & ANATOMY

Audit: Strong influencer facial structure with zero plastic smoothing. Burgundy nails, gold band ring, and open-mouth geometry preserved. Minor residual eyelash density from source, but recognizable likeness achieved.

Test Set 4: Hyper-Personalized Grounded Prompt (Outputs 7 & 8)

```
# Recreate This Exact Opening Frame With the Selected Influencer
**SPECIFIC RECONSTRUCTION TASK**: Take the original vertical photograph in Image 1, showing a surprised woman
wearing a light-wash denim jacket in a bright white room, with her burgundy-nailed hand raised to her chin.
Replace that woman's visible facial identity with the specific brunette influencer shown in Images 2–5...
**THE ACTUAL INFLUENCER IDENTITY — THE MOST IMPORTANT REQUIREMENT**: Naturally full, dark, softly shaped
eyebrows; hazel-brown eyes; distinct nose bridge and tip; tapered jawline; fine natural freckles around nose;
deep brunette wavy hair...
**SPECIFIC SOURCE IDENTITY FEATURES TO REPLACE**: The original performer has prominently styled eyelashes,
different brow/eye structure, and lighter-highlighted hair. Do not retain these features...
**PRECISE FOREGROUND PRESERVATION**: Preserve all five fingers, burgundy nail polish, gold-toned band ring
```



OUTPUT 7: BENCHMARK GOLD STANDARD

Audit: Flawless execution. The influencer is immediately recognizable: authentic hazel eyes, natural delicate freckles, realistic brow geometry, and brunette hair. The denim jacket, warm sconces, burgundy nails, and ring are 100% pixel-faithful.



OUTPUT 8: MAXIMUM CONSISTENCY

Audit: Zero identity leakage from source creator. The open-mouth surprised reaction is anatomically adapted to the influencer's mouth structure without distortion. Complete caption cleanup with seamless fabric reconstruction.

Benchmark Scorecard & Technical Synthesis

Quantitative Findings & Implementation Strategy for UGCPilot

ENGINEERING SPEC

Comparative Performance Scorecard

Evaluation Dimension	App Standardized	Custom Prompt 1	Refined Prompt 2	Hyper-Personalized
Influencer Resemblance	2.5 / 10 Poor	5.5 / 10 Fair	8.0 / 10 Good	**9.8 / 10** Ideal
Source Identity Rejection	1.0 / 10 Severe Leak	4.5 / 10 Moderate	8.0 / 10 High	**9.7 / 10** Complete
Expression & Emotion Fidelity	5.0 / 10 Unstable	8.5 / 10 Good	9.0 / 10 Good	**9.9 / 10** Flawless
Hand, Nails & Prop Occlusion	4.0 / 10 Drift	8.5 / 10 Good	9.0 / 10 Good	**9.8 / 10** Exact
Lighting & Background Match	8.5 / 10 Good	9.0 / 10 Good	9.2 / 10 Good	**9.8 / 10** Seamless
Caption & Overlay Cleanliness	8.0 / 10 Good	9.0 / 10 Good	9.2 / 10 Good	**9.9 / 10** Seamless
Overall Benchmark Score	**4.8 / 10 (FAIL)**	**7.4 / 10 (PARTIAL)**	**8.7 / 10 (PASS)**	**9.8 / 10 (EXCELLENT)**

Key Engineering Findings & Architectural Recommendations

1. Why the Hyper-Personalized Prompt Triumphed

- **Disentangled Biometric Descriptors:** Diffusion models cannot parse abstract terms like "Identity Authority" effectively. Calling out "*hazel-brown eyes, natural dark eyebrows, fine natural freckles around the nose, wavy brunette hair*" provides the cross-attention layers with concrete generative targets.
- **Negative Semantic Exclusion:** Specifically instructing the model to "*replace the prominently styled eyelashes, different eyebrow and eye structure, and highlighted hair*" directly breaks the model's tendency to copy source eye makeup.
- **Concrete Prop Anchoring:** Explicitly describing "*burgundy nail polish, long nails, gold-toned band ring, and chin proximity*" prevented the model from hallucinating generic hands or covering the mouth (as happened in Output 2).

2. Production Strategy: The Dynamic Hybrid Engine

Rather than forcing a rigid one-size-fits-all prompt across the application, the app should adopt a **Layered Dynamic Generator**:

- **Core Immutable Rules:** Subject-first role order, aspect ratio, clean fabric reconstruction under captions, zero replacement logos.
- **Auto-Extracted Scene Slot (from ugc-frame-analysis):** Injects detected wardrobe, props, hand positions, nail colors, and background lighting.
- **Auto-Extracted Identity Slot:** Injects the selected influencer profile's distinctive biometric traits (eye color, brow density, freckles, hair type).